Measuring Privacy-Preserving Computation in One Place: MPC, FHE, ZKP and Their Hybrids

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Abstract

Privacy-preserving computation is not one field but four—secure multi-party computation (MPC), fully homomorphic encryption (FHE), garbled circuits, and zero-knowledge proofs (ZKP)—each with its own literature, benchmarks and accounting conventions. A practitioner choosing between them, or a reviewer assessing a claim that one beats another, must assemble numbers from different papers, measured on different hardware, over different networks, counting different things. Cross-paradigm comparison is therefore largely *citation assembly* rather than measurement.

We take a different route. We survey the paradigms through a single artifact, Baru, in which one source program runs on all of them—a plain interpreter, three-party replicated-secret-sharing MPC, garbled circuits, BFV/BGV/CKKS/TFHE homomorphic encryption, a Binius64 SNARK, and *combinations* in which values cross between representations mid-computation without being exposed. Every backend is driven through the same transport, charged by the same byte and round accounting, and checked bit-exactly against the same plaintext reference. A network cost model is validated against a real socket to within 13.6 % worst case, so a claim about a wide-area deployment is computed from measured counts rather than asserted.

Several widely repeated intuitions survive re-measurement; several do not. Among the results that only a single-harness study can produce: the FHE-versus-MPC crossover is a property of *the baseline's preprocessing* rather than of either cryptosystem, and disappears once MPC is given somewhat-homomorphic preprocessing; active security on the nonlinear surface of private inference costs 2.63× bytes but only 1.06× rounds; and the choice between a garbled and a `wrap3`-style comparison protocol has a measured crossover in bandwidth that moves with round-trip time. We also report our own withdrawn claims, including a ZKP backend that proved nothing and a bit-decomposition that leaked its inputs, because a study that reports only what worked is not reporting.

1 The comparability problem

Ask how fast private inference is and the literature answers in incompatible units. Gazelle [24], Delphi [32] and Cheetah [22] report two-party hybrid systems in which the server holds the model in plaintext. Falcon [38] reports three-party replicated sharing on a LAN. CKKS [11] papers report amortized throughput over batched slots. SNARK papers report prover time for a circuit that a different paper would express with a different constraint count. Each is internally rigorous. Together they do not compose.

Four obstacles recur.

1. **Different hardware.** A 5× difference between two systems is unfalsifiable when one ran on a server-class Xeon and the other on a laptop.

2. **Different accounting.** "Communication" may mean payload bytes, serialized bytes, or bytes per party. "Rounds" may mean protocol rounds, messages, or interactive calls. These differ by more than the effects being measured.

3. **Different security models.** Semi-honest and malicious figures appear in adjacent columns of secondary literature without the distinction surviving the citation.

4. **Different functions.** Two ReLU costs may be for different bit widths, different fixed-point scales, or—in the worst case we found in our own work—subtly different functions that nobody checked against a common reference.

A survey assembled from such numbers inherits all four. The alternative we pursue is to re-measure the paradigms in one harness where the obstacles are controlled by construction: the same machine, the same accounting code, machine-checked security levels, and a bit-exactness assertion against a shared plaintext reference before any cost is quoted.

The cost of this approach is equally clear and we state it up front. Our implementations are research-quality Rust, not the tuned C++ of the systems we discuss. **Absolute figures here are not competitive with optimized implementations and should not be read as such.** What the single harness buys is *internally valid ratios*: comparisons in which the confounds above are held fixed. Where we compare against published numbers we say so, and treat the comparison as weaker evidence than our internal ones.

Contributions.

- A **systematization** of four paradigms organized by what they actually cost, with the cost dimensions (rounds, bytes, compute, preprocessing) separated rather than collapsed into one "performance" number (Section 3).
- A **single-artifact measurement study**: roughly 72,000 lines of Rust in which one program runs on every paradigm, with a validated network model and uniform accounting (Sections 4 and 6).
- **Cross-paradigm findings** not visible from any single paradigm's literature (Section 7), including two crossovers whose location depends on a parameter the original claims left implicit.
- A **methodology for honest benchmarking** (Section 6), with the record of what it caught in our own work (Section 9).

2 Scope

In scope. Semi-honest and malicious three-party MPC over rings; garbled circuits in the honest-majority setting; RLWE-based FHE (BFV [21], BGV [7], CKKS [11]) and TFHE [13]; FHE↔MPC scheme switching; SNARKs over 64-bit words (Binius64 [18]); and the supporting primitives these need—oblivious transfer, private set intersection, private information retrieval, and Beaver-triple [4] preprocessing from both OT and somewhat-homomorphic encryption (SHE).

Paradigm	Secret from	Expensive in	Cheap in	Rounds
Secret-sharing MPC	computing parties	**rounds**, bandwidth	local arithmetic	O(depth)
Garbled circuits	computing parties	**bandwidth**	rounds	$O(1)$
FHE	computing server	**compute**, ct. size	rounds (none)	none
ZK proofs	the verifier	**prover compute**	verifier, proof size	none

Table 1: The paradigms organized by cost structure rather than security property. Three trade-offs follow directly, and drive most of this paper.

Out of scope. Differential privacy and federated learning (a different threat model: they protect an *aggregate*, not a computation); trusted execution environments (hardware trust assumption); ORAM (surveyed but not implemented, see Section 11); and transformer nonlinearities such as softmax and LayerNorm, which need exponential, reciprocal and inverse-square-root subprotocols we do not have.

Application focus. We use private neural-network inference as the running workload, because it is the one application that exercises every paradigm: dense linear algebra (where FHE packing shines), comparisons (where MPC and garbled circuits compete), and verifiable computation (where SNARKs apply). Where a result is specific to this workload we say so.

3 Four paradigms, four cost structures

The paradigms are usually taxonomized by security property. For a practitioner choosing among them, the more useful axis is *what each is expensive in* (Table 1).

MPC and garbled circuits trade rounds against bytes. This is the classic Yao [41]-versus-GMW trade. Neither dominates: the winner depends on round-trip time and bandwidth, and the boundary is an empirical question. Section 7.3 measures it.

FHE and MPC trade compute against communication. FHE sends a lot once and computes hard; MPC sends little, many times. On a fast LAN, MPC usually wins; on a thin pipe, the balance shifts. This framing omits *preprocessing*, and Section 7.1 shows the omission is what makes most published FHE-versus-MPC crossovers fragile.

ZK is orthogonal; conflating it with the others is a category error. MPC and FHE keep data secret *from the parties computing on it*. A SNARK proves a computation was performed correctly while hiding the witness *from a verifier*—the prover holds everything in the clear. A deployment needing both confidentiality and integrity pays for both. We keep them in separate tables for this reason, and our benchmark prints the caveat alongside its numbers so they cannot travel without it.

3.1 The hybrid, and why it exists

Neural-network inference alternates linear layers with nonlinearities. FHE is good at the former and bad at the latter—comparison is not a low-degree polynomial. MPC is the reverse: comparison costs a bit decomposition, but a linear layer costs one round regardless of width.

Linear layers under FHE, nonlinearities under MPC, with a *scheme switch* between them, is therefore the natural design, and is what Gazelle [24], Delphi [32] and Cheetah [22] all instantiate. Baru implements it and adds two things (Section 5.4): the weights are encrypted too, and the switch is used as a *noise refresh*, which removes bootstrapping from deep networks entirely.

4 The Baru artifact

4.1 One program, many backends

Baru is a Lisp-like language whose interpreter manipulates secret values only through opaque handles and a fixed set of primitive operations. This is the Arithmetic Black Box abstraction of the SPDZ line [16, 17]: an ideal functionality $\mathcal{F}_{\mathsf{ABB}}$ storing values by handle and exposing `add/mul/lt/share_public/reveal`. A backend implements those primitives; the evaluator never inspects a representation.

```
; private ReLU on one neuron -- every value is secret-shared
(let ((w :sint8 -1) (x :sint8 3) (b :sint8 0))
  (let ((z (+ (* w x) b)))
    (reconstruct (* z (> z 0)))))          ; => 0

; a value computed under MPC crosses into FHE without being revealed
(let ((keys (fhe-keygen)) (x (share 6 32)) (y (share -7 32)))
  (fhe-dec keys (share->fhe keys (* x y))))   ; => -42
```

The consequence for a survey is direct: *the same source text* is the workload for every paradigm, so a cross-paradigm comparison cannot be confounded by two implementations computing subtly different things. This is not hypothetical; Section 9 records a case where it caught exactly that.

4.2 Composability

Because every gadget (`relu`, `max2`, `maxpool2d`, `argmax`, `dense`, `conv2d`) is a pure composition of $\mathcal{F}_{\mathsf{ABB}}$ primitives, universal composability [9] gives security of any network built from them for free, given the primitives. Architecture search therefore needs no new security argument per architecture, and the research question becomes "which architecture is *fastest* under the cost model?" rather than "is this one secure?"

4.3 What is implemented

Table 2 summarizes. All figures below are from the maintainer's machine (Apple M1 Pro, 8 cores), release build, unless noted.

5 The paradigms, surveyed and measured

5.1 Secret-sharing MPC

Baru's MPC layer is ABY3-style [33]: $x = s_0 + s_1 + s_2 \bmod 2^{128}$, with party i holding the replicated pair (s_i, s_{i+1}). Addition and scalar multiplication are local. Multiplication is **triple-free**—each party sends one ring element to one neighbour, with correlated randomness regenerated locally from a one-time pairwise PRF key exchange. There is no dealer and no preprocessing phase.

The engineering lesson this layer taught, four separate times, generalizes beyond the artifact.

Layer	Contents
MPC	ABY3 [33] replicated shares over $\mathbb{Z}_{2^{128}}$; **triple-free** multiplication from pairwise PRFs; bit decomposition (carry-save $3 \to 2$ plus Kogge–Stone [28]); batched primitives; **malicious security with abort** (≤ 1 of 3 corrupt), following Chida et al. [12] in the ring adaptation of Eerikson et al. [20]; in-process and TCP transports
Garbled	Half-gates [42], free-XOR [29], fixed-key AES; comparator and fused ReLU $\circ$ rescale circuits; OT-free label delivery in the three-party honest-majority setting
Falcon	Π_{PC} private compare over $\mathbb{Z}_p$, Π_{WA} `wrap3`, Π_{DReLU}, Π_{SS}, Π_{Maxpool}, exact truncation—a reproduction of Falcon [38] inside the same harness
FHE	BFV [21] and BGV [7] (depth-one, no evaluation keys), SEAL CKKS [35, 11], TFHE-rs [37, 13]; multi-modular NTT with Shoup butterflies; CRT slot packing; Cheetah-style [22] convolution packing
Hybrid	MPC$\leftrightarrow$FHE switching both directions; re-encryption as a bootstrapping alternative; replicated threshold-key switch [3]
ZKP	Expression-to-circuit compiler; real Binius64 [18] prover and verifier
Compiler	Backend-aware circuit IR, rewrite rules, cost models, protocol registry (dealer / Shamir / Semi2k [25] / LowGear [26]), per-operation protocol assignment
Primitives	Chou–Orlandi [14] and IKNP [23] OT, DH-PSI, single-server BFV PIR

Table 2: What the artifact implements. Every row is exercised by at least one benchmark, and every engine attests its security level (Section 6.4).

Workload	bytes	interactive calls	wall clock (WAN, 25 ms)
Comparisons, n=32, 32-bit	**2.63**×	**1.06**×	**1.26**×
Dot product, n=32, 32-bit	2.47×	1.75×	2.80×

Table 3: Semi-honest ABY3 versus malicious-with-abort, same workload, same transport. Active security costs bytes, not rounds.

Finding. The arithmetic was never the bottleneck; the round structure was. Each time a segment-level measurement said "MPC is slow", a per-phase breakdown said "one unbatched loop inside it is slow".

Batched forms cost *the same number of rounds for any batch width.* Adopting them moved the CNN's MPC segment from 40.8 s to 6.1 s (6.7×), with messages falling from 29,211 to 5,591. A reader should take from this that published MPC costs are highly sensitive to batching discipline, and that comparing two systems' round counts without checking whether both batch is comparing implementations, not protocols.

Cost of active security. Verified 2026-08-01 (Table 3).

The comparison row is the one that matters for inference, since every ReLU, max-pool and argmax reduces to a comparison.

Finding. Active security costs bytes, not rounds: the MAC doubles what is sent but batches into the same round.

The dot-product row is a small-workload artifact—the semi-honest leg needs only four interactive calls, so a fixed pair-pool cost amortizes against nothing—and we report it rather than hide it.

A second result generalizes. The statistical slack that ring-based verification normally charges as a widening to $\mathbb{Z}_{2^{k+s}}$ [15] is *already paid for* when the implementation shares over a ring much wider

than its values. Q16.16 values in a 128-bit ring leave **59 bits** of statistical security at no widening cost.

5.2 Garbled circuits

In the *three-party honest-majority* setting, garbled-circuit input labels are delivered by the co-garbler with no oblivious transfer. Since OT is the expensive part of two-party garbling, this changes the economics substantially, and is why garbling is competitive here at all.

Measured head-to-head against bit decomposition through the identical transport (1,352 comparisons, 20-bit): `a2b` needs 9 rounds and 5.94 MB; the garbled circuit needs 1 round and 1.69 MB. Wall clock favours garbling 4.2× on a LAN and 4.9× on a 20 ms WAN.

A correction to our own earlier claim. That 4.2–4.9× is the *garbling primitive in isolation*. The deployable three-party protocol, including the $Y2A$ return leg that converts the boolean output back to an arithmetic share, is 1.4–1.7×, worth 1.9× on the CNN's MPC segment. We record both, because the gap between them is exactly what gets lost in citation.

5.3 Fully homomorphic encryption

Baru uses the cheapest FHE primitive that exists: one ciphertext×ciphertext multiplication, *no relinearization keys, no bootstrapping, no evaluation keys*, decrypting the degree-2 product directly under s^2. Parameters are $n = 4096$, $q = 2^{100}$, $t = 2^{20}$ for BFV—inside the ≈109-bit $\log q$ ceiling for 128-bit security at that dimension [1].

Two packing results are worth surveying.

- **Reversed-coefficient packing** makes a whole matrix–vector product one multiplication. We find this is *already optimal for dense layers*: a degree-n product admits at most n/in separable outputs, so $\lceil \text{out} \cdot \text{in}/n \rceil$ multiplications is the limit. Gazelle's SIMD packing [24] matches the throughput but requires rotations and Galois keys, forfeiting the no-evaluation-keys property.
- **Cheetah-style convolution packing** [22] makes a whole output channel one multiplication, independent of image size. Measured A/B at 28 × 28: 1,352 multiplications become **2**, and FHE time falls from 77.5 s to 264 ms.

Finding. BGV is indistinguishable from BFV at identical parameters (repeated runs put the gap at +0.5%, +0.5%, −0.1%). Where the message sits in the ciphertext, high bits or low, is not where the cost is.

On arithmetic, our multi-modular NTT—NTT-friendly primes used as a computational device and CRT-reconstructed, because a power-of-two q admits no direct NTT—is 1.8× faster than NTL+GMP [36] at every size after moving its butterfly to Shoup-preconditioned multiplication. NTL is retained as a *differential oracle* rather than a dependency, which is what made rewriting the hottest code in the repository safe.

5.4 The hybrid and the no-bootstrap result

The construction's central observation is that the scheme switch performed for a ReLU *also refreshes the ciphertext*. After switching back, each party freshly encrypts its share component and the evaluator sums; the sum of fresh encryptions is a fresh encryption. The FHE noise budget therefore

depth	exact?	total	fhe	switch	mpc	our boots.	CKKS-competitor boots.
2	✓	145 ms	20	59	54	**0**	0
8	✓	539 ms	81	239	210	**0**	3
32	✓	2,119 ms	333	958	820	**0**	15

Table 4: Deep private inference, width 16, weights *and* activations encrypted, real ABY3 protocol, BFV at $n = 4096$, $q = 2^{100}$. Linear in depth and bit-exact at every depth; the exactness is the evidence for the noise claim, since accumulated noise would make deep outputs diverge.

width	constraints	prove	verify	proof bytes
1	14	6.3 ms	1.0 ms	18,544
64	589	8.0 ms	1.3 ms	47,808
784	**7,159**	**32.3** ms	**4.8** ms	**111,712**

Table 5: Constraints grow linearly (≈ 9 per input); proof size does not—a 286× larger circuit yields a 6× larger proof. Width 784 is the MNIST input dimension.

resets at every layer boundary, and a network of arbitrary depth runs with **zero FHE bootstraps** at cost linear in depth, on an engine supporting only a *single* multiplication (Table 4).

This construction also encrypts the **weights**. That is a stronger privacy model than Gazelle, Delphi and Cheetah, whose server holds the model in plaintext.

5.5 Zero-knowledge proofs

Proving a Q16.16 quantized neuron, $\mathrm{ReLU}((\sum_i w_i x_i) \gg 16 + b)$, with public weights and private inputs, on a real Binius64 prover (Table 5).

Two implementation findings generalize to anyone building arithmetic circuits over 64-bit words. First, comparison gates are **unsigned**, so signed comparison must flip sign bits first; without it $\mathrm{ReLU}(-1)$ returns -1—a circuit perfectly sound about the wrong function. Second, and sharper:

Finding. Division cannot be checked the obvious way. A circuit cannot divide, so the prover supplies quotient and remainder as witnesses; but checking $a = bq + r$ with $|r| < |b|$ is *unsound*, because circuit multiplication wraps mod 2^{64}. For $a = 1, b = 3$ a cheating prover sets $r = 0$ and $q = 3^{-1} \bmod 2^{64}$ and both checks pass. Requiring the 128-bit product's high word to be the sign extension of the low word closes it.

We drive that forgery through the real prover as a test.

5.6 Preprocessing: OT versus SHE

Beaver-triple preprocessing is where the FHE-versus-MPC comparison is usually decided, and it is usually decided implicitly (Table 6).

A comparability caveat the literature routinely drops—and which we dropped ourselves. A ≈60-bit prime field does *not* represent every 64-bit value, so Table 6 compares a 61-bit field triple against a 64-bit ring triple. At **equal 64-bit width** LowGear needs two CRT limbs, giving 384 B/triple against OT's 3,074—an **8.0**× communication advantage, not the 16× a naive reading suggests. Our own experimental record quoted 16.3×; we correct it here. This is precisely the error class a single-harness study is positioned to catch, and it is the reason Section 6 exists.

route	domain	B/triple	triples/s	rounds
LowGear (SHE) [26]	$\mathbb{F}_p$, 60-bit prime	**192**	97,304	2
Semi2k (OT) [25]	$\mathbb{Z}_{2^{64}}$	3,074	2,681	3

Table 6: Two parties, semi-honest. **The two rows are over different domains**; see the caveat below for the apples-to-apples figure.

The measured OT cost also validates the analytic model: 3,074 measured against 3,136 predicted, a 2 % error.

6 Methodology

This is the section we would most like other artifacts to adopt.

6.1 Identical transport, uniform accounting

Every protocol—arithmetic MPC, garbled, Falcon, hybrid—runs over the same transport stack, wrapped in counting and latency decorators that observe traffic without touching protocol code. Both legs of any comparison pay real serialization, real message passing and real byte counting. There is no measurement asymmetry in which one side is simulated and the other is not.

This matters more than it sounds. The in-process simulator *is not the protocol*: its share multiplication reconstructs and reshares locally, which is free and insecure, and it understated the pure-MPC baseline by ≈2.7× (6.7 ms simulated against 18 ms real). Every MPC number in this study is the real wire protocol, with three party threads exchanging serialized messages.

6.2 A validated network model

The cost model charges rounds × RTT + bytes/bandwidth, letting a benchmark "move" parties between continents without deploying anything. It is **validated against a real loopback TCP socket**: worst error 13.6 %, under 1 % through the middle of both sweeps, with the link's constants measured rather than assumed. Cost really is linear in rounds and linear in bytes with no missing cross-term, which is what licenses computing wall-clock from measured counts.

Building this validation found two harness bugs that had been silently distorting results: per-message latency timers could *reorder* back-to-back messages, and the bandwidth model did not serialize bulk transfers cumulatively, *undercharging exactly the byte-heavy phases* the crossover experiments are about. Both now have regression tests. We mention this because a cost model never checked against a socket should be assumed to have bugs of this kind.

The honest limitation: the constants are loopback's (RTT 23 µs), not a WAN's, so a wide-area deployment needs its own two constants.

6.3 Bit-exactness as a gate on quoting cost

Finding. A cost comparison between routes computing different things is meaningless.

Every backend is therefore checked against a common plaintext reference before any number is reported, and the benchmarks refuse to present a clean table when a cell disagrees. The encrypted CNN reproduces the plaintext prediction on 600 of 600 embedded medical images, run twice—before

width	dealer (lower bound)	Semi2k (OT)	**LowGear (SHE)**	hybrid/BFV
64	600.6 ms	694.5 ms	**606.2** ms	692.1 ms
1024	603.0 ms	942.8 ms	**623.4** ms	692.1 ms
4096	610.9 ms	1.74 s	**678.4** ms	733.7 ms

Table 7: One private layer with both operands secret plus a ReLU; WAN, two parties, preprocessing charged per triple *consumed*.

and after two optimizations—with both passes agreeing on all 600. That is what makes those optimizations *exact* rather than merely close.

6.4 Machine-checked security attestations

Every engine attests the security level it actually implements, in one auditable table, and asking a session for more than an engine attests is an *error rather than a silent downgrade*. Exactly one engine attests malicious security. The editing rule is that an attestation records what the code *does*, never what the roadmap intends: raising one is the last step of an upgrade, after the cheating-detection tests pass.

We consider this the most transferable idea in the artifact. Security level is the dimension most often lost when numbers move between papers, and making it a machine-checked property of running code—rather than prose in a limitations section—is cheap to implement and hard to get wrong by accident.

6.5 Recording withdrawn results

Negative, superseded and withdrawn results are kept in the experimental record alongside the wins, with the reasoning that produced them. Section 9 summarizes what this caught.

7 Cross-paradigm findings

These are the results that motivated the single-harness approach, in the sense that no single paradigm's literature could produce them.

7.1 The FHE-versus-MPC crossover belongs to the baseline's preprocessing

The standard claim is that FHE overtakes MPC past some layer width, because MPC preprocessing grows linearly while a packed ciphertext is flat. Baru's cost model predicts it, and measurement confirms it—*against an OT-preprocessed baseline* (Table 7).

Against **SHE-preprocessed** MPC the crossover is gone in this range entirely. LowGear tracks the trusted-dealer lower bound within 5 ms at width 64.

Finding. The honest statement is not "the hybrid beats dishonest-majority MPC past width 64" but "the hybrid beats *OT-preprocessed* dishonest-majority MPC past width 64, and loses to SHE-preprocessed MPC at every width measured." The mechanism survives—preprocessing volume decides it—but the boundary is a property of the baseline, not of either cryptosystem.

A second boundary moves with the **threat model**. Under an honest majority (ABY3, multiplication from correlated randomness, zero preprocessing) the layer stays on MPC at *every* width. Under a dishonest majority with OT preprocessing, FHE takes it past ≈256 multiplications. This

RTT	protocol	1000	100	40	30	20	10	5 Mbit s^{-1}
0 ms	gc	**638**	**3227**	**6554**	**8511**	12,357	23,915	46,869
	falcon	1,279	3,528	6,750	8,583	**12167**	**22828**	**43794**
20 ms	gc	**2411**	**4839**	**8395**	**10292**	**14174**	25,594	48,659
	falcon	3,873	6,276	9,585	11,393	14,998	**25543**	**46850**

Table 8: Milliseconds per encrypted MedMNIST diagnosis; winner in bold. Both routes compute the same diagnosis bit-exactly, from one shared session, so only the protocol and the network vary.

explains a structural fact about the literature that is otherwise puzzling: *every hybrid FHE+MPC inference system is two-party.* Gazelle, Delphi and Cheetah sit precisely in the regime where the model says the hybrid pays.

7.2 Protocol assignment can be derived, not annotated

Given measured costs for each backend, a search over per-operation-kind assignments returns `mul` → fhe, `cmp-lt` → binary for a 256-wide encrypted layer with a comparison—that is, it *derives* the Gazelle/Delphi/Cheetah architecture rather than having it annotated. The prior work on protocol assignment (HyCC [8] → Silph [10] → SING [5]) is **MPC-only**; extending the cost model to span MPC *and* FHE is what makes the hybrid architecture an output of the search rather than an input.

Two structural non-linearities make this objective non-separable, which is worth noting for anyone tempted to formulate it as an integer linear program: a dependence level costs the *maximum* over its instructions, not their sum; and conversions *batch per level*, so a crossing's cost depends on whether another crossing already happens there.

7.3 The nonlinearity crossover is in bandwidth, and moves with RTT

Baru implements three interchangeable comparison protocols—batched bit decomposition, garbled circuits, and Falcon's `wrap3`-based Π_{DReLU} [38]—behind one interface, verified to compute the same function bit-exactly. This makes the trade of Section 3 directly measurable on a real workload rather than argued (Table 8).

Falcon—the lowest-bandwidth route—wins below ≈20 Mbit s^{-1} at RTT 0 and below ≈10 Mbit s^{-1} at RTT 20 ms. Higher RTT shifts the crossover *down*, shrinking the region where it wins, because it sends more messages. The mechanism checks out quantitatively: Falcon starts ≈641 ms behind on compute and pays 67 extra messages, against which it saves 1.6 MB; break-even is predicted at 20.0 and 6.5 Mbit s^{-1} against 20–30 and 10–15 measured.

The boundary is a band, not a line. Around 20–30 Mbit s^{-1} the winner flips between repeat runs (0.1–0.3% spread). No ordering inside that band should be quoted; the finding is that the protocols are *indistinguishable* there, which is what a crossover is.

Finding. Given a deployment's RTT and bandwidth, one should *pick* the nonlinearity protocol rather than fixing it. Systems that hardwire one choice are optimal only in part of the deployment space.

	PneumoniaMNIST	OrganAMNIST
task	chest X-ray, 2 classes	abdominal CT, 11 organs
integer accuracy, full test set	84.78 % (624)	81.18 % (17,778)
95 % Wilson CI [39]	[81.7, 87.4]	[80.6, 81.8]
balanced accuracy / AUC	80.64 % / **0.9314**	80.49 % / **0.9590**
sensitivity / specificity	97.18 % / **64.10** %	—
published ResNet-18 (28×28)	0.854 / 0.944	0.935 / 0.997
majority-class baseline	62.50 %	18.48 %
encrypted = plaintext	**300/300**	**300/300**
FHE bootstraps	0	0
online latency / diagnosis	**268** ms	291 ms

Table 9: Private diagnosis on real biomedical images, image *and* weights encrypted, only the class index revealed.

8 Case study: private medical image diagnosis

MedMNIST v2 [40] distributes twelve biomedical datasets at 28×28—exactly MNIST's [30] shape—so a private-inference pipeline runs on them unchanged, and the artifact stops being about handwritten digits (Table 9).

Three points a privacy venue should care about.

Privacy costs nothing in accuracy, and this is shown rather than asserted. The integer model matches its float parent to within 0.2 points, and the encrypted pipeline reproduces the plaintext diagnosis bit-exactly on every one of 600 images.

Accuracy alone misreads a medical classifier. PneumoniaMNIST's test set is ≈60% positive, and this model earns its 84.78 % largely by *over-calling disease*: 97.2 % sensitivity against 64.1 % specificity. Balanced accuracy is 80.64 %. None of this is visible while only accuracy is reported, and none of it is a cryptographic result—but a paper claiming private diagnosis has to report it.

The accuracy ceiling is the packing, not the crypto. We initially attributed the model's size to the FHE noise budget. That was wrong. Three limits bind and noise is the third: spatially, 3×3 valid convolution plus pooling takes $28 \to 13 \to 5 \to 1$, admitting at most three convolutions; by *packing*, $2 \cdot \mathrm{in}_c \cdot H \cdot W \lesssim n$ caps the first convolution at 12 channels before the second's input passes $n = 4096$. That is a packing constraint that relaxes at $n = 8192$, not a limit of the construction. Depth, meanwhile, really is free.

9 What the methodology caught in our own work

A study's credibility rests partly on what its authors were willing to withdraw. Each item below was found by the discipline of Section 6 and is recorded in the artifact rather than quietly fixed.

A ZKP backend that proved nothing. Until 2026-07-31 the shipped backend's `verify` never checked that a witness satisfied any constraint— anyone holding the public circuit could forge an accepting proof. It is retained, unused, so a soundness test can keep demonstrating precisely what it failed to do. **Every ZKP number predating that date is withdrawn.**

A privacy break in bit decomposition. The previous `a2b` masked its input with a value derived from a *public* lockstep counter, so the input was recoverable from the transcript by any observer. Everything built on it— ReLU, max-pool, argmax—leaked its activations. The advertised "no extra communication per bit" was a *symptom*: the carry chain was free because every operand was already public. Fixing it cost 4.4× MPC time and ≈130× bandwidth, later recovered by triple-free multiplication and batching.

An insecure FHE scheme, removed rather than caveated. The fast engine was a scale-invariant NTRU [31, 6] port. Analysis showed it insecure at the parameters the construction needs even at depth one [2, 27, 19]: the large LTV secret raises the fatigue point to $\approx 2^{59}$, but a single multiply with usable t forces $q \gtrsim 2^{84}$. It was deleted, not kept as a "fast reference"; every result transferred bit-exact and the FHE column got slower.

A selective-failure channel that passes every functional test. Attaching MACs to bit-decomposition leaves by multiplication is unsound: a corrupt party injects δ, the first downstream gate carries MAC error δy with y a secret bit, so the run aborts when $y = 1$ and completes when $y = 0$.

Finding. Deferring the abort does not close this. Batching hides *when* an abort happens, not *whether* it happens; those are different channels and only the first is closed by batching. We had asserted the opposite before building it.

Comparison errors of framing. Pricing FHE's *linear* part against MPC's *whole layer* (omitting the ReLU that FHE cannot do at all); computing a crossover against a trusted-dealer baseline (asking whether FHE beats free preprocessing); and the cross-domain triple comparison corrected in Section 5.6.

Honest negatives retained. PIR loses to downloading the database by ≈13× with no crossover in N. Hybrid-TFHE does not pay. Faster ring arithmetic does *not* bring FHE close to MPC. Per-layer bit-width tuning was measured at ≈10% and not built. A fused garbled max-of-four was measured as a 1.68× *loss* and not built.

10 Threats to validity

Implementation quality. Research Rust, not tuned C++. Absolute figures are not competitive; ratios measured within the harness are the defensible output. Where we compare against published numbers—for example our sign extraction at ≈0.043 ms against Falcon's optimized C++—the gap is implementation, not protocol, and we say so.

Single machine. All figures from one Apple M1 Pro. Cross-machine variance is uncharacterized. The network model is validated on loopback, so WAN constants are modelled.

The simulated network is a model. Validated to 13.6 % worst case, but it is still rounds × RTT + bytes/bandwidth, not a real WAN with congestion, jitter or packet loss.

Two disclosed cryptographic gaps. The Falcon backend's preprocessing is currently *dealt from a common seed and is not secure*—all three parties can reconstruct the mask. Its online protocol and measured costs are real; its offline phase is not. FHE threshold *key generation* is dealer-based. Both are recorded on the code and in the attestation table rather than only in prose.

Workload breadth. Findings are anchored on feed-forward CNN inference at 28×28. Transformers, larger images, and non-inference workloads are not covered.

Not an adversarial evaluation. Cheating-detection tests drive real attacks over the transport for the malicious ABY3 layer, but this is not a security audit, and no formal proofs are machine-checked.

11 Open problems

1. **A CIFAR-scale private CNN** needs SIMD-slot packing; coefficient packing bounds image size. The composition and security story is independent of it.
2. **Transformer nonlinearities** (softmax, LayerNorm) need exponential, reciprocal and inverse-square-root subprotocols.
3. **Malicious security at the FHE boundary.** Replication cannot catch a *malformed input ciphertext*—both honest co-holders compute on it faithfully and agree. This sharpens the positioning: *ZK is needed only at the input boundary, never for the evaluation*, where single-server verifiable FHE needs it over the whole computation. The boundary proof is not implemented.
4. **Secret-index array access**, priced by the cost model, before ORAM. CirC [34] implements exactly the two cases either side of that line and does not implement ORAM at all.
5. **Per-instruction protocol assignment**, versus the current per-operation-kind granularity.
6. **Cross-machine and real-WAN validation** of the cost model's constants.

12 Conclusion

Cross-paradigm performance claims in privacy-preserving computation are not currently falsifiable, because the numbers come from incommensurable sources. We have described a harness in which they are, and reported what changes when the paradigms are re-measured under uniform accounting: an FHE-versus-MPC crossover that belongs to the baseline's preprocessing rather than to either cryptosystem; active security that costs bytes but not rounds; and a nonlinearity choice that should be made per deployment rather than fixed by the system designer. Three claims did not survive the re-measurement, two of them our own.

The transferable part is not the artifact but the discipline: identical transports for both legs of any comparison, a cost model checked against a real socket, bit-exactness as a precondition for quoting cost, and security level as a machine-checked property of running code rather than prose. Each is cheap to adopt, and each caught something here that would otherwise have been published.

Reproducibility

Single Rust crate, no GPU. `cargo test` covers correctness; every `bench-*` subcommand emits `-json`. All numbers are produced by these commands; none are hand-entered.

Result	Command
Deep inference, zero bootstraps	`bench-deep-nn -depths 2,4,8,16,32`
Private diagnosis (Table 9)	`bench-medmnist -dataset pneumonia`
Nonlinearity crossover (Table 8)	`bench-medmnist -sweep`
Cost of active security (Table 3)	`bench-malicious -n 32 -bits 32`
Triples: SHE vs OT (Table 6)	`bench-triples`
FHE↔MPC crossover (Table 7)	`bench-layer`
Cost model vs real socket (Section 6.2)	`bench-net`
Proving a quantized neuron (Table 5)	`bench-zkp-layer -widths 1,...,784`
Falcon vs `a2b` vs garbled	`bench-falcon`

References

[1] Martin Albrecht et al. Homomorphic encryption security standard. HomomorphicEncryption.org, Toronto, Canada, 2018.

[2] Martin R. Albrecht, Shi Bai, and Léo Ducas. A subfield lattice attack on overstretched NTRU assumptions. In *CRYPTO*, pages 153–178, 2016.

[3] Gilad Asharov, Abhishek Jain, Adriana López-Alt, Eran Tromer, Vinod Vaikuntanathan, and Daniel Wichs. Multiparty computation with low communication, computation and interaction via threshold FHE. In *EUROCRYPT*, pages 483–501, 2012.

[4] Donald Beaver. Efficient multiparty protocols using circuit randomization. In *CRYPTO*, pages 420–432, 1991.

[5] Blüml, Huppert, Khayata, Schmidt, and Schneider. SING: Hybrid MPC protocol assignment with graph neural networks. Cryptology ePrint Archive, Paper 2026/1237; to appear, USENIX Security 2026, 2026. `https://eprint.iacr.org/2026/1237`.

[6] Joppe W. Bos, Kristin Lauter, Jake Loftus, and Michael Naehrig. Improved security for a ring-based fully homomorphic encryption scheme. In *IMA International Conference on Cryptography and Coding*, pages 45–64, 2013.

[7] Zvika Brakerski, Craig Gentry, and Vinod Vaikuntanathan. (leveled) fully homomorphic encryption without bootstrapping. *ACM Transactions on Computation Theory*, 6(3), 2014.

[8] Niklas Büscher, Daniel Demmler, Stefan Katzenbeisser, David Kretzmer, and Thomas Schneider. HyCC: Compilation of hybrid protocols for practical secure computation. In *ACM SIGSAC Conference on Computer and Communications Security (CCS)*, pages 847–861, 2018.

[9] Ran Canetti. Universally composable security: A new paradigm for cryptographic protocols. In *42nd IEEE Symposium on Foundations of Computer Science (FOCS)*, pages 136–145, 2001.

[10] Edward Chen, Jinhao Zhu, Alex Ozdemir, Riad S. Wahby, Fraser Brown, and Wenting Zheng. Silph: A framework for scalable and accurate generation of hybrid MPC protocols. In *IEEE Symposium on Security and Privacy (S&P)*, pages 848–863, 2023.

[11] Jung Hee Cheon, Andrey Kim, Miran Kim, and Yongsoo Song. Homomorphic encryption for arithmetic of approximate numbers. In *ASIACRYPT*, pages 409–437, 2017.

[12] Koji Chida, Daniel Genkin, Koki Hamada, Dai Ikarashi, Ryo Kikuchi, Yehuda Lindell, and Ariel Nof. Fast large-scale honest-majority MPC for malicious adversaries. In *Advances in Cryptology – CRYPTO 2018*, pages 34–64, 2018.

[13] Ilaria Chillotti, Nicolas Gama, Mariya Georgieva, and Malika Izabachène. TFHE: Fast fully homomorphic encryption over the torus. *Journal of Cryptology*, 33(1):34–91, 2020.

[14] Tung Chou and Claudio Orlandi. The simplest protocol for oblivious transfer. In *Progress in Cryptology – LATINCRYPT 2015*, pages 40–58, 2015.

[15] Ronald Cramer, Ivan Damgård, Daniel Escudero, Peter Scholl, and Chaoping Xing. SPDZ_{2^k}: Efficient MPC mod 2^k for dishonest majority. In *Advances in Cryptology – CRYPTO 2018*, pages 769–798, 2018.

[16] Ivan Damgård and Jesper Buus Nielsen. Universally composable efficient multiparty computation from threshold homomorphic encryption. In *CRYPTO*, pages 247–264, 2003.

[17] Ivan Damgård, Valerio Pastro, Nigel P. Smart, and Sarah Zakarias. Multiparty computation from somewhat homomorphic encryption. In *CRYPTO*, pages 643–662, 2012.

[18] Benjamin E. Diamond and Jim Posen. Succinct arguments over towers of binary fields. Cryptology ePrint Archive, Paper 2023/1784, 2023. `https://eprint.iacr.org/2023/1784`.

[19] Léo Ducas and Wessel P. J. van Woerden. NTRU fatigue: How stretched is overstretched? In *ASIACRYPT*, pages 3–32, 2021.

[20] Hendrik Eerikson, Marcel Keller, Claudio Orlandi, Pille Pullonen, Joonas Puura, and Mark Simkin. Use your brain! arithmetic 3PC for any modulus with active security. In *1st Conference on Information-Theoretic Cryptography (ITC)*, pages 5:1–5:24, 2020.

[21] Junfeng Fan and Frederik Vercauteren. Somewhat practical fully homomorphic encryption. Cryptology ePrint Archive, Paper 2012/144, 2012. `https://eprint.iacr.org/2012/144`.

[22] Zhicong Huang, Wen jie Lu, Cheng Hong, and Jiansheng Ding. Cheetah: Lean and fast secure two-party deep neural network inference. In *31st USENIX Security Symposium*, pages 809–826, 2022.

[23] Yuval Ishai, Joe Kilian, Kobbi Nissim, and Erez Petrank. Extending oblivious transfers efficiently. In *Advances in Cryptology – CRYPTO 2003*, pages 145–161, 2003.

[24] Chiraag Juvekar, Vinod Vaikuntanathan, and Anantha Chandrakasan. GAZELLE: A low latency framework for secure neural network inference. In *27th USENIX Security Symposium*, pages 1651–1669, 2018.

[25] Marcel Keller. MP-SPDZ: A versatile framework for multi-party computation. In *ACM SIGSAC Conference on Computer and Communications Security (CCS)*, pages 1575–1590, 2020.

[26] Marcel Keller, Valerio Pastro, and Dragos Rotaru. Overdrive: Making SPDZ great again. In *Advances in Cryptology – EUROCRYPT 2018*, pages 158–189, 2018.

[27] Paul Kirchner and Pierre-Alain Fouque. Revisiting lattice attacks on overstretched NTRU parameters. In *EUROCRYPT*, pages 3–26, 2017.

[28] Peter M. Kogge and Harold S. Stone. A parallel algorithm for the efficient solution of a general class of recurrence equations. *IEEE Transactions on Computers*, C-22(8):786–793, 1973.

[29] Vladimir Kolesnikov and Thomas Schneider. Improved garbled circuit: Free XOR gates and applications. In *35th International Colloquium on Automata, Languages and Programming (ICALP)*, pages 486–498, 2008.

[30] Yann LeCun, Léon Bottou, Yoshua Bengio, and Patrick Haffner. Gradient-based learning applied to document recognition. *Proceedings of the IEEE*, 86(11):2278–2324, 1998.

[31] Adriana López-Alt, Eran Tromer, and Vinod Vaikuntanathan. On-the-fly multiparty computation on the cloud via multikey fully homomorphic encryption. In *44th ACM Symposium on Theory of Computing (STOC)*, pages 1219–1234, 2012.

[32] Pratyush Mishra, Ryan Lehmkuhl, Akshayaram Srinivasan, Wenting Zheng, and Raluca Ada Popa. DELPHI: A cryptographic inference service for neural networks. In *29th USENIX Security Symposium*, pages 2505–2522, 2020.

[33] Payman Mohassel and Peter Rindal. ABY3: A mixed protocol framework for machine learning. In *ACM CCS*, pages 35–52, 2018.

[34] Alex Ozdemir, Fraser Brown, and Riad S. Wahby. CirC: Compiler infrastructure for proof systems, software verification, and more. In *IEEE Symposium on Security and Privacy (S&P)*, pages 2248–2266, 2022.

[35] Microsoft SEAL (release 4.1). `https://github.com/Microsoft/SEAL`, 2023. Microsoft Research, Redmond, WA.

[36] Victor Shoup. NTL: A library for doing number theory. `https://libntl.org/`, 2021.

[37] TFHE-rs: A pure rust implementation of the TFHE scheme. `https://github.com/zama-ai/tfhe-rs`, 2024. Zama.

[38] Sameer Wagh, Shruti Tople, Fabrice Benhamouda, Eyal Kushilevitz, Prateek Mittal, and Tal Rabin. FALCON: Honest-majority maliciously secure framework for private deep learning. *Proceedings on Privacy Enhancing Technologies*, 2021(1):188–208, 2021.

[39] Edwin B. Wilson. Probable inference, the law of succession, and statistical inference. *Journal of the American Statistical Association*, 22(158):209–212, 1927.

[40] Jiancheng Yang, Rui Shi, Donglai Wei, Zequan Liu, Lin Zhao, Bilian Ke, Hanspeter Pfister, and Bingbing Ni. MedMNIST v2 – a large-scale lightweight benchmark for 2D and 3D biomedical image classification. *Scientific Data*, 10(1):41, 2023.

[41] Andrew Chi-Chih Yao. How to generate and exchange secrets. In *27th IEEE Symposium on Foundations of Computer Science (FOCS)*, pages 162–167, 1986.

[42] Samee Zahur, Mike Rosulek, and David Evans. Two halves make a whole: Reducing data transfer in garbled circuits using half gates. In *EUROCRYPT*, pages 220–250, 2015.